WORKSHEET

Chapter 5: Circular motion

1 Each of the following gives the revolutions per minute completed by a circular disc and the radius of the disc. In each case, calculate the angular velocity in rad s^{-1} and the speed of a point on the circumference of the disc in m s^{-1}.

a 100 rpm with radius of 2 m

b 250 rpm with radius of 1.25 m

c 500 rpm with radius of 3 m

d 1000 rpm with radius of 4.5 m

e 450 rpm with radius of 1.85 m

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f 20 rpm with radius of 5.5 m

2 A gymnast on a high bar swings through two (clockwise) revolutions in a time of 1.90 s. Find the angular velocity of the gymnast.

3 An aeroplane propeller is rotating with uniform angular speed of 1 800 rpm. The blades of the propeller are 1 m long. Determine the linear speed of a point:

a 1 m from the centre of the propeller

b 50 cm from the centre.

4 A point on a circle rotates through $\frac{3\pi}{4}$ radians in 3 s. Give the angular velocity of the point.

5 A disc with a radius of 13 cm turns with an angular velocity of 3 radians per second. Find the distance travelled by a point on edge of the disc in 1 minute.

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6 A disc is turning at 45 rpm. If the distance from the centre of the disc to a point on the edge of the disc is 3 cm, find the angular velocity and the linear velocity of the point.

7 Calculate the angular velocity of a point on the circumference of a 0.3 m radius wheel when the wheel travels at 15.0 m s^{-1}.

8 Determine the magnitude of the acceleration of a car following a curve of radius 500 m at a speed of 25.0 m s^{-1}.

9 Determine the length of a person's arm if she can swing an object around five times in a circle at 2.72 m s^{-1} in 7.5 s.

10 Find the acceleration of an object that moves around a circular path with a radius of 10 m once every 14 s.

11 Find the acceleration of a car driving at 60 km h^{-1} in a circular track that is 120 m across.

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12 The angular velocity of an aeroplane propeller is increased from 1 800 rpm to 2 200 rpm in 10 s. Determine its angular acceleration.

13 An object sweeps out a central angle of $\frac{\pi}{3}$ radians in 0.5 s as it moves around a circle of radius 3 m. Find its angular velocity and speed over that time period.

14 An object travels a distance of 35 m in 2.7 s as it moves around a circle of radius 2 m. Find its angular velocity and speed over that time period.

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Answers

1 **a** angular velocity $= 10.47$ rad s^{-1}; speed $= 20.94$ m s^{-1}

b angular velocity $= 26.18$ rad s^{-1}; speed $= 32.72$ m s^{-1}

c angular velocity $= 52.36$ rad s^{-1}; speed $= 157.08$ m s^{-1}

d angular velocity $= 104.72$ rad s^{-1}; speed $= 471.24$ m s^{-1}

e angular velocity $= 47.12$ rad s^{-1}; speed $= 87.18$ m s^{-1}

f angular velocity $= 2.09$ rad s^{-1}; speed $= 11.52$ m s^{-1}

2 -6.63 rad s^{-1}

3 **a** 188.4 m s^{-1}

b 94.2 m s^{-1}

4 $\frac{\pi}{4}$ rad s^{-1}

5 23.40 m

6 Angular velocity $= \frac{3\pi}{2}$ rad s^{-1}; v $= 14.13$ cm s^{-1}

7 $\omega = 50$ rad s^{-1}.

8 a $= 1.25$ m s^{-2}

9 0.65 m

10 2.01 m s^{-2}

11 4.63 m s^{-2}

12 4.19 rad s^{-2}

13 $\omega = \frac{2\pi}{3}$ rad s^{-1}; $v = 2\pi$ m s^{-1}.

14 $\omega = 6.48$ rad s^{-1}; $v = 12.96$ m s^{-1}.

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